A joint count–moment estimate on the left probability range

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The third absolute moment can decrease when noise is added to a Bernoulli law with parameter below 1/3. Estimating that decrease separately from the own-count loss is wasteful: both estimates involve the same two conditional third moments. Combining them first produces the positive reserve in Lemma 2. This is enough to extend the noise range from 1/500 to 1/100 on the whole interval $p \in [1/4, 2/5]$. No subdivision of that probability interval or Bernoulli reserve is used.

Theorem 1 (LEFT-01)**.** *Let $n \geq 42$, $1/4 \leq p \leq 2/5$, $q = 1 - p$, and $Y = B + U$, where B is Bernoulli with parameter p and $\mathbb{E}(U \mid B) = 0$. Suppose the conditional laws U_j satisfy*

$$-A \leq U_0 \leq a_0, \qquad -a_1 \leq U_1 \leq A, \qquad A = 7/10, \quad a_0, a_1 \geq 0, \quad a_0 + a_1 \leq 17/250.$$

Put $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $\ell = (v+s)^{-1/2}$, $X = \ell(Y-p)$, $\rho = \mathbb{E}|X|^3$, and $\rho_B = (1-2v)/\sqrt{v}$. Assume $0 < \lambda \leq 1/100$ and $|H| \leq 46/25$. Let S_n be a sum of n independent copies of X. Choose $k \in \mathbb{Z}$ and u by

$$np + H/\ell = k + u, \qquad -109/500 \leq u \leq 391/500, \qquad \delta = k - np.$$

Such an anchor always exists, and it is interior: $1 \leq k \leq n-1$. For every $819/2000 \leq C \leq 41/100$, writing F_B for the distribution function of $\mathrm{Bin}(n,p)$, one has

$$\begin{aligned}\Delta_C = \sqrt{n}\{\mathbb{P}(S_n \leq H) - \Phi(H/\sqrt{n})\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B < -\lambda/100.\end{aligned}$$

The constant $C_E = (3 + \sqrt{10})/(6\sqrt{2\pi})$ lies strictly between the two stated bounds. The external strict Bernoulli theorem therefore excludes a nonnegative excess at target C_E under these hypotheses. At $\lambda = 0$ the law is itself Bernoulli. The comparison theorem has no contact or predecessor premise. Those premises remain necessary for the separate acquisition of its support and threshold bounds. Reflected events require applying the acquisition and comparison to the reflected law; a bound on one signed branch is not an unqualified two-sided result.

1 Condition on the actual count

Throughout set

$$l = -109/500, \quad h = 391/500, \quad b = 1/20, \quad U_* = 1/100, \quad v_* = 3/16, \quad D = 17/10.$$

The phase interval has width one, so $k = \lfloor np + H/\ell - l \rfloor$ is an admissible choice. Let W_j be the sum of j independent copies of U_1 and $n-j$ independent copies of U_0, all independent. Write $\pi_j = \binom{n}{j} p^j q^{n-j}$, set $m = \sqrt{n}\pi_k$, and define

$$f(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}.$$

$$O = \mathbb{P}(W_k > u) + f(u), \qquad T_{\mathrm{in}} = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \le u - i).$$

Exact conditioning gives

$$\begin{aligned}\mathbb{P}(S_n \le H) - F_B(k) = & -\sum_{i=0}^{k} \pi_{k-i}\mathbb{P}(W_{k-i} > u + i) \\ & + \sum_{i=1}^{n-k} \pi_{k+i}\mathbb{P}(W_{k+i} \le u - i).\end{aligned}$$

Thus, if $\pi_{k+i}/\pi_k \le R$ on the legal incoming counts,

$$\Delta_C \le -m(O - RT_{\mathrm{in}}) + R_N + C(\rho_B - \rho), \qquad R_N = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\}. \quad (1)$$

All outgoing events are strict; incoming events are closed. No continuity assumption or removal of threshold atoms enters this identity.

2 Uniform finite normalization

We give the constants explicitly because the small improvement in normalization helps the left range. Here $v_* \le v \le 6/25$ and $\sqrt{v+s} < 491/1000$. Consequently

$$|\delta| < D, \qquad |\delta| < 1 \ (|u| \le b), \qquad |\delta| < 101/100 \ (|u| \le 1/10).$$

The segment from p to k/n is contained in $(1/5, 9/20)$ globally, and in $(9/40, 17/40)$ in the last two cases. This proves that k is interior. On these segments, respectively,

	$\lvert D'''(r\Vert p)\rvert$	$\lvert\{\log(r(1-r))\}'\rvert$	$1/(12r(1-r))$
global	24	4	53/100
near	19	16/5	12/25

are strict upper bounds. They follow from $D'''(r\|p) = -(1-2r)/(r^2(1-r)^2)$ and the endpoint bounds for $r(1-r)$. Finite Stirling,

$$\log(j!) = (j + 1/2)\log j - j + \tfrac{1}{2}\log(2\pi) + r_j, \qquad 0 < r_j < 1/(12j),$$

and Taylor's theorem yield

$$m = \frac{\phi(0)}{\sqrt{v}} \exp\{-\delta^2/(2nv) + e\}, \qquad n|e| < Z(D_0), \quad (2)$$

where, when $|\delta| \le D_0$,

$$Z_g(D_0) = 2D_0 + 53/100 + 4D_0^3/42, \qquad Z_n(D_0) = (8/5)D_0 + 12/25 + (19/6)D_0^3/42.$$

For example the entropy error before multiplication by n is bounded by $19D_0^3/(6n^2)$ on the near segment; the prefactor and factorial errors are bounded by $(8/5)D_0/n$ and $12/(25n)$. The finite factorial proof is retained in `manuscript/arxiv-v1/appendices/count-comparison.tex`, line 131.

At a fixed count,

$$\log f'(z) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta + z)^2}{2n(v+s)} - e.$$

Since $1/(2v_*) + D^2/(84v_*^2) < 4$, for $|z| \le b_0$ this gives

$$n|\log f'(z)| \le 4\lambda + \frac{2D_0b_0 + b_0^2}{2v_*} + Z(D_0).$$

Apply $e^{-x} \ge 1 - x$ and $e^x \le (1 - x)^{-1}$, using $(D_0, b_0) = (D, h), (1, b), (101/100, 1/10)$ and the corresponding Z. In (2), use $\sqrt{v} < 49/100$ and $19943/50000 < \phi(0) < 2/5$. The resulting bounds are

	$\lvert u\rvert \le b$	$l \le u \le h$
m lower bound	18/25	57/100
f' lower bound	47/50	2/3
f' upper bound	213/200	3/2
π_{k+i}/π_k upper bound	53/50	9/5

(3)

We also have $f' > 93/100$ for $|u| \le 1/10$. All bounds hold on the whole interval of integration defining f. At a binomial mode the displacement is at most one. Equation (2) gives $\sqrt{n}\max_j \pi_j < 1$, proving the global ratio bound in (3). In the inner interval the first ratio is at most $1 + p/(nv) < 53/50$; all subsequent ratios are at most one, since $p - \delta < 7/5 < 2$.

Write $s_j = \mathbb{E}U_j^2$. Because $|s_1 - s_0| \le 4s$, $\mathbb{E}W_j^2 = \lambda + (j - np)(s_1 - s_0)$. Hence for $t = \mathbb{E}W_k^2$,

$$(19/21)\lambda \le t \le (23/21)\lambda \quad (|u| \le b), \qquad t \le (7/6)\lambda \quad \text{globally.} \tag{4}$$

For the incoming counts $\mathbb{E}W_{k+i}^2 \le iV\lambda$, where $V = 6/5$ in the inner interval and $V = 4/3$ globally.

The normal repair satisfies $R_N \le (5/2)\delta_+ s$, since $0.2/v_*^{3/2} < 5/2$. The inward caps fix the signs in $|Y - p|$. Using $U_0^3 \le a_0U_0^2$, $U_1^3 \ge -a_1U_1^2$, and $(1 + x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$ gives

$$(v + s)^{3/2}(\rho - \rho_B) \ge s\{3\min(p, q) - \tfrac{3}{2}(1 - 2v) - 17/250 - 3(1 - 2v)s/(8v)\}.$$

The expression in braces is at least $-B_-$, where

$$B_- = \tfrac{3}{2}(1 - 2v_*) + 17/250 - \tfrac{3}{4} + 3(1 - 2v_*)U_*/(8 \cdot 42v_*).$$

Rational squaring verifies $(41/100)B_-/v_*^{3/2} < 4/3$. Thus the separate repairs in (1) are bounded by $4s$ in the inner interval and $6s$ globally. We shall improve that bound near the negative phase origin by combining the moments first.

3 The joint reserve

Set $\alpha = 27/50$ and $M = 5\alpha^4$. The polynomial

$$h_\alpha(x) = \frac{-x^4 + 3\alpha^2x^2 + 2\alpha^3x}{M} \le \mathbf{1}_{(0,\infty)}(x)$$

is a strict-event minorant: its numerator is $-x(x + \alpha)^2(x - 2\alpha)$, and its positive maximum is $(9 + 6\sqrt{3})\alpha^4/4 < 5\alpha^4$. For $r \ge 0$ define the familiar separate-moment lower expression

$$L(r, t) = \frac{(3\alpha^2 - A^2 - 6r^2)t - 3t^2 - 4Art - r^4 + 3\alpha^2r^2 + 2\alpha^3r}{M}. \tag{5}$$

Lemma 2. *Under the theorem's hypotheses, if $u = -r$ and $0 \le r \le b$, then*

$$m\mathbb{P}(W_k > -r) + C(\rho - \rho_B) > mL(r, t) + \tfrac{7}{4}s.$$

Proof. Put $N_0 = n - k$, $N_1 = k$, $T_j = N_j s_j$, and $\mu_j = \mathbb{E}U_j^3$. The convex function $x^2 - \kappa x$ attains its maximum at an endpoint, so

$$\mathbb{E}U_0^4 - \kappa\mu_0 \le \max\{A^2 + \kappa A, a_0^2 - \kappa a_0\}s_0,$$

and the corresponding coefficient for U_1 is $\max\{A^2 - \kappa A, a_1^2 + \kappa a_1\}$. Centering and independence give

$$\mathbb{E}W_k^3 = N_0\mu_0 + N_1\mu_1, \quad \mathbb{E}W_k^4 = \sum \mathbb{E}U_i^4 + 3t^2 - 3\sum(\mathbb{E}U_i^2)^2.$$

The exact marginal identity is

$$\rho = \ell^3\{v(1-2v) + 3v(s_0+s_1) + p\mu_1 - q\mu_0\}.$$

In $m\mathbb{E}h_\alpha(W_k + r) + C(\rho - \rho_B)$, the two cubic prices are therefore

$$\kappa_0 = -4r - \gamma q/N_0, \qquad \kappa_1 = -4r + \gamma p/N_1, \qquad \gamma = C\ell^3 M/m.$$

The outer endpoint coefficients apply if $\kappa_0 \ge -(A - a_0)$ and $\kappa_1 \le A - a_1$. Indeed $\ell^3 < 25/2$, $m > 18/25$, $q/N_0 < 1/38$, $p/N_1 < 1/38$, and the exact inequality

$$4b + \frac{(41/100)(25/2)M}{(18/25)38} < A - 17/250$$

proves both guards. The reciprocal count bounds follow from $|\delta| < 1$, $p \ge 1/4$ and $q \ge 3/5$.

Consequently the fourth/cubic contribution is at least

$$-A^2 t + 4Ar(T_0 - T_1) + \gamma A(qs_0 + ps_1) - 3t^2.$$

Here the favorable $3\sum(\mathbb{E}U_i^2)^2$ was discarded. Bound $T_0 - T_1 \ge -t$ and $v(s_0 + s_1) \ge \min(p, q)s$. Applying the same power inequality to the Bernoulli normalization now gives

$$\begin{aligned} m\mathbb{P}(W_k > -r) + C(\rho - \rho_B) \ge{}& mL(r,t) \\ &+ C\ell^3 s\{A + 3\min(p,q) - \tfrac{3}{2}(1-2v) - 3(1-2v)s/(8v)\}. \end{aligned}$$

The braces are at least

$$B_+ = A + \tfrac{3}{4} - \tfrac{3}{2}(1 - 2v_*) - 3(1-2v_*)U_*/(8 \cdot 42v_*).$$

Finally $(819/2000)B_+ > (7/4)(491/1000)^3$ and $\sqrt{v+s} < 491/1000$. This proves the claim. Notice the reserve is proportional to $s = \lambda/n$, not to λ. □

4 The own count at small phase

The usual absolute-moment bounds $|\mathbb{E}W_k^3| \le At$ and $\mathbb{E}W_k^4 \le A^2 t + 3t^2$ are enough outside the use of Lemma 2. After shifting the quartic by σr, adding cr, dividing by $t > 0$, and setting $r = zt$, their lower expression is

$$\begin{aligned} F_{\sigma,c}(z,t) = \frac{3\alpha^2 - A^2 - 3t}{M} + \left(\frac{2\sigma}{5\alpha} + c - \frac{4At}{M}\right) z \\ + \frac{3\alpha^2 t - 6t^2}{M} z^2 - \frac{t^3 z^4}{M}. \end{aligned}$$

Let $T_* = (23/21)U_*$ and $z_0 = 3/2$. The following checks are on entire rectangles, not a sampled mesh:

$$F_{1,-213/200} > 1/4, \qquad F_{-1,47/50} > 4/5 \quad (0 \le z \le z_0,\ 0 \le t \le T_*).$$

For the first polynomial, its z derivative is at most $2/(5\alpha) - 213/200 + 6\alpha^2 T_* z_0/M < 0$. It is concave in t, so its minimum is among $(z_0, 0), (z_0, T_*)$. For the second polynomial its z derivative is bounded below by $47/50 - 2/(5\alpha) - 4AT_*/M - 4T_*^3 z_0^3/M > 0$; the quadratic derivative omitted here is nonnegative since $3\alpha^2 - 6T_* > 0$. Its minimum is therefore at $(0, T_*)$. All three corner values exceed the stated bounds.

For negative phase beyond $r = z_0 t$ use Cantelli:

$$O \geq G_t(r) := \frac{r^2}{t + r^2} - \frac{213}{200} r > t/2, \qquad z_0 t \leq r \leq b.$$

To verify the last inequality, G_t' first increases and then decreases. It is positive at $z_0 t$, so the minimum on this interval is at an endpoint. At the switch, $G_t(z_0 t)/t = z_0^2/(1 + z_0^2 t) - (213/200) z_0 > 1/2$. At b, $G_t(b) - t/2 > 0$. Both tests are decreasing in t and are checked at T_*; also $z_0 T_* < b$. For positive phase after $r = z_0 t$, use $O \geq f(r) \geq (47/50) r > (4/5) t$.

Combining the first quartic bound with Lemma 2 and $R_N \leq (5/2)s$ gives a repair of at most $(3/4)s$ when $-z_0 t \leq u \leq 0$. For the remaining negative inner phases use the Cantelli own bound $t/2$ and repair $4s$. For positive inner phases the own bound is $4t/5$ with repair $4s$.

5 The incoming counts, including the infinite tail

We retain the quadratic-hinge comparison of Pinelis, *On the Bennett–Hoeffding inequality*, arXiv:0902.4058, Proposition 2.8 and Corollary 2.11. For independent centered summands bounded above by A, with total variance at most V_0, it gives

$$\mathbb{P}(Z \geq x) \leq \frac{\mathbb{E}(P_\mu - j)_+^2}{(x/A + \mu - j)^2}, \qquad \mu = V_0/A^2, \quad j < x/A + \mu,$$

where P_μ is Poisson. This follows by applying its moment inequality to the quadratic hinge at $A(j - \mu)$ and then Markov's inequality; increasing the variance upper bound is allowed. We apply it to $-W_{k+i}$, preserving the closed event and using the same two conditional marginals.

Set $\mu = V\lambda/A^2$. For $u \leq h_0 < 1 - A$, the first incoming count is bounded by

$$Q(\mu, h_0) = \frac{\mu^2 (1/2 + \mu/6)}{((1 - h_0)/A - 1 + \mu)^2}.$$

Indeed the hinge is $j = 1$ and $\mathbb{E}(P_\mu - 1)_+^2 = \mu^2 - \mu + 1 - e^{-\mu} \leq \mu^2/2 + \mu^3/6$. For integer $i \geq 2$, $(P - i)_+^2 \leq \binom{P}{i+1}$ pointwise. For $i = 2$ this is $(r - 1)(r - 2) \geq 0$ at integer $r = P - i \geq 1$; increasing i increases the binomial coefficient at fixed r. The Poisson numerator is consequently at most $(i\mu)^{i+1}/(i + 1)!$. For $j = 2$ or 3 put

$$b_j(h_0) = 1/A - 1 - h_0/(jA), \qquad J_2(\mu, h_0) = \frac{\mu^3}{3(b_2(h_0) + \mu)^2 (1 - 3\mu)},$$

$$J_3(\mu, h_0) = \frac{3\mu^4}{8(b_3(h_0) + \mu)^2 (1 - 3\mu)}.$$

If $h_0 \geq 0$, $b_j(h_0) > 0$ and $3\mu < 1$, these bound the sum of incoming probabilities from count $i = j$ onwards. The hinge at count i has denominator at least $i(b_j(h_0) + \mu)$, and the successive coefficient ratio is $[i/(i + 2)](1 + 1/i)^i < e < 3$. This proves the geometric sum for all counts, without truncation. In particular $b_3(h) = 59/1050 > 0$, whereas the second hinge cannot be used at h. At large positive phase the first two counts are instead bounded by Cantelli.

All the displayed tail bounds increase with λ. Also $Q(c\lambda, h_0)/\lambda$ increases while $c\lambda \leq (1 - h_0)/A - 1$, and $J_2(c\lambda, h_0)/\lambda$ increases when $b_2(h_0) > 0$. The first assertion follows by differentiating $\mu(1/2 + \mu/6)/(a + \mu)^2$; its derivative has numerator $a/2 + \mu(a/3 - 1/2) > 0$ for $0 \leq \mu \leq a$. Thus these estimates are uniform down to arbitrarily small positive noise.

6 Closing the four phase intervals

All remaining arithmetic consists of the following rational comparisons. Let

$$\mu_I = (6/5)U_*/A^2, \qquad \mu_G = (4/3)U_*/A^2, \quad R_I = 53/50, \quad R_G = 9/5.$$

In the inner interval define

$$T_- = Q(\mu_I, 0) + J_2(\mu_I, 0), \qquad T_+ = Q(\mu_I, b) + J_2(\mu_I, b).$$

For the three inner cases the expression

$$\frac{18}{25}\left(c\frac{19}{21} - R_I T_\pm/U_*\right) - a/42 - 1/100$$

is strictly positive with

$$(c, T_\pm, a) = (1/4, T_-, 3/4), \quad (1/2, T_-, 4), \quad (4/5, T_+, 4).$$

The bracket itself is positive in each case. Equations (1)–(4), the joint estimate and the tail monotonicity therefore give $\Delta_C < -\lambda/100$ throughout $[-b, b]$. The first of these three margins is $314105373/15227387000 > 0$.

For negative outer phase set $T_G = 7U_*/6$ and

$$B_-^{\text{out}} = \min_{r\in\{b,-l\}}\left\{\frac{r^2}{T_G + r^2} - \frac{3}{2}r\right\} - R_G\{Q(\mu_G, 0) + J_2(\mu_G, 0)\}.$$

The minimum really is at the endpoints: the derivative of the expression in braces first increases and then decreases and is positive at b. This handles $u \in [l, -b]$. For the positive intervals put $t_G = 4U_*/3$ and

$$K(u) = \frac{t_G}{t_G + (1-u)^2} + \frac{2t_G}{2t_G + (2-u)^2}.$$

On $[b, 1/10]$ use the bracket

$$B_+^{\text{near}} = (93/100)b - R_G\{K(1/10) + J_3(\mu_G, 1/10)\}.$$

On $[1/10, h]$ use

$$B_+^{\text{far}} = \min_{u\in\{1/10,h\}}\{(2/3)u - R_G K(u)\} - R_G J_3(\mu_G, h).$$

Here the endpoint minimum is valid by concavity, since $3(1-h)^2 > t_G$ and $3(2-h)^2 > 2t_G$. For each of these three positive brackets, the exact check is

$$(57/100)B - 6U_*/42 - U_*/100 > 0.$$

The respective margins are

$$\frac{12211588349}{224360220000}, \qquad \frac{5230759451}{6254703000000}, \qquad \frac{32120174181617}{2611470546230000}.$$

They prove the required inequality because $\lambda \le U_*$. The four closed intervals

$$[l, -b], \qquad [-b, b], \qquad [b, 1/10], \qquad [1/10, h]$$

cover the entire phase domain. The inner switch uses the actual variance t, and both sides include its endpoint. This completes the proof.

Verification and scope

`leftcount-check.py` verifies all 88 strict scalar gates with `Fraction`, including every branch guard, positive denominator, phase join and final margin. Machin's formula with alternating remainders supplies rational enclosures for π, so the stated normal and C_E bounds are also checked. The checker rejects missing, duplicated, reordered and gapped phase lists. Other negative controls reject the unguarded endpoint price, the invalid second hinge at h, and the attempt to close this 1/100 argument after discarding its joint reserve.

The unfavorable moment sign is not hypothetical: for $p = 13/50$, $U_1 = 0$, and $U_0 = \pm 1/1000$ equiprobably, rational squaring gives $\rho < \rho_B$. This refutes deleting that correction, not LEFT-01. The separate actual-law script checks the new coupled inequality at exact atoms and at tiny positive noise. Such finite examples test implementation and conventions; they do not establish the uniform theorem.

For the acquired witnesses the left-panel heat/ray strips of v11 imply $U_0 > -.6$ and $U_1 < .7$, which imply the amplitude assumption used here. Their contact, nonnegative-excess and sharp-predecessor premises remain ancestors, as do the raw threshold and direct geometry bounds. This result replaces the count exclusion only for $p \in [.25, .4]$ and $0 < \lambda \leq .01$ at the stated target range. It neither proves a finite base through 41 nor excludes the remaining larger-noise region. It does not delete an existing certificate or alter a previous theorem.